README

Zambia Wildlife Biodiversity Dashboard

An Excel data analysis project that cleans, models and visualises wildlife occurrence records for Zambia (1962 to 2026) from the Global Biodiversity Information Facility (GBIF), using Power Query, Power Pivot (DAX), PivotTables and an interactive dashboard.

📖 Project Overview

GBIF occurrence data is rich but messy: many taxonomy columns (phylum, class, order, family) have blanks, dates are stored as timestamps, and the download contains dozens of columns that are not needed for analysis. This project turns the raw download into a clean, analysis-ready data model and answers questions about what has been recorded in Zambia, when, and where.

Questions answered

1. **Time trends (1962 to 2026):** How have records changed per year? Which months have the most sightings?
2. **Location:** Which provinces have the most records and the most species?
3. Which class has the most records?

🗂️ Data Source

Item	Detail
Source	GBIF occurrence download
Coverage	Zambia, 1962 to 2026
Download date	21 October 2026
Citation	GBIF.org 21 October 2026 GBIF Occurrence Download https://doi.org/10.15468/dl.5njh3q
Basis of record	All records are human observations

🔄 Workflow

```
Raw GBIF download
      |
      ▼
Power Query: clean text, standardise blanks, flag rows with missing taxonomy
      |
      ▼
GBIF Species Lookup: match unique scientific names to get phylum, class, order, family
      |
      ▼
Power Query: merge results back, fill blanks only, fix dates, keep analysis columns
      |
      ▼
Power Pivot: data model, DAX measures, Decade column, Years table
      |
      ▼
PivotTables, PivotCharts and slicers  ⟶  Dashboard
```

🧹 Data Cleaning (Power Query)

- Trimmed text and converted empty cells to proper nulls
- Flagged records with a blank phylum, class, order or family
- Built a list of unique scientific names for those records and matched them with GBIF Species Lookup
- Merged the results back (left join on `scientificName`) and filled **only** the blank values, keeping original values untouched
- Labelled names GBIF could not match as `Unknown` so they stay visible instead of silently dropping out
- Converted `eventDate` from a timestamp (`T` and `Z`) to a proper date
- Reduced the dataset to the columns needed for analysis

Columns kept: gbifID, kingdom, phylum, class, order, family, genus, species, taxonRank, scientificName, stateProvince, occurrenceStatus, eventDate, day, month, year, basisOfRecord

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🧮 Data Model (Power Pivot)

- One fact table (`Wildlife`)
- A calculated column, `Decade`
- DAX measures that handle blanks correctly: Records, Unique Species, Unique Genera, Unique Families, Unique Orders, Unique Classes, Unique Phyla, Records with Species, Records per Species, Records Incl Zero

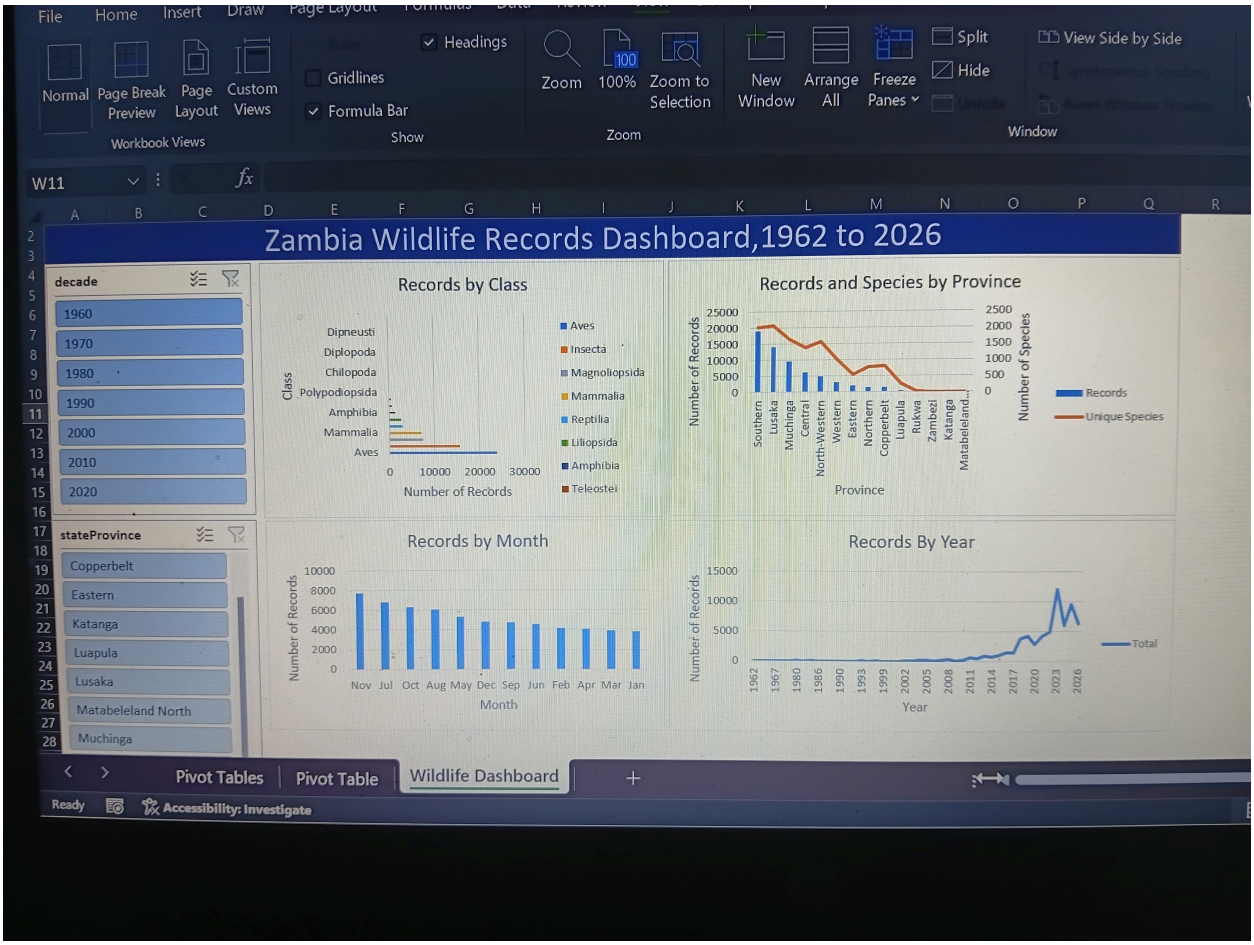
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📊 Dashboard

The dashboard combines 4 charts and 2 slicers (Decade and province).

#	Chart	Source
1	Records per year	PivotChart (line)
2	Records by month	PivotChart (column)
3	Records and species by province	PivotChart (combo)
4	Records by class	PivotChart (bar)

Dashboard



Dashboard

🔍 Key Findings

- **Most recorded class:** Aves with 23962 records.
- **Recording trend:** The records per year show an upward trend from trend ,with 2023 recording the highest number of records at approximately 12,000.
- **Seasonality:** November, July and October are the top 3 months with the most sightings.
- **Provinces:** Southern province has the most records, and Lusaka province has the most species

⚠️ Limitations

- **Recording effort, not abundance:** all records are human observations, so counts reflect where and when people looked, not how common a species is. Each record is one occurrence entry, not one animal.
- **Partial decades:** the 1960s cover 1962 to 1969 and the 2020s cover 2020 to 2026.
- **Missing and unmatched values:** records without a species name are left out of species counts. Names GBIF could not match are blank.

🛠️ Tools

- Microsoft Excel (Power Query, Power Pivot, PivotTables, PivotCharts, slicers)
- DAX
- GBIF Species Lookup
- Git and GitHub

📂 Repository Structure

```
zambia-wildlife-biodiversity-dashboard/
│
├── data/
│   ├── raw/                    # Original GBIF download
│   └── cleaned/                # Cleaned dataset used by the workbook
│
├── workbook/
│   └── Zambia_Wildlife_Dashboard.xlsx
│
├── docs/
│   ├── dashboard.png           # Dashboard screenshot
│   ├── power_query_steps.md    # Cleaning steps and M code
│   └── dax_measures.md         # Data model and DAX measures
│
├── README.md
└── LICENSE                     # MIT
```

▶️ Getting Started

1. Download or clone this repository.
2. Open `workbook/Zambia_Wildlife_Dashboard.xlsx` in desktop Excel for Windows or Microsoft 365. Power Pivot is needed to edit the data model.
3. Go to the dashboard sheet and use the slicers to filter by decade, class or province.
4. To see how it was built, read the Power Query and DAX documents in `docs/`.
5. To refresh from your own data, update the file path in the Power Query source step, then choose **Data > Refresh All**

🌟 About Me

Hi, I'm Chama Mulubwa. I built this project to practise data cleaning, data modelling and dashboard design with real biodiversity data from Zambia.

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- GitHub: https://github.com/mulubwac20-hub